

- 1 A ball of mass 10 g is dropped from a height and falls through air. The variation with time t of the speed of the ball v is shown in Fig. 1.1

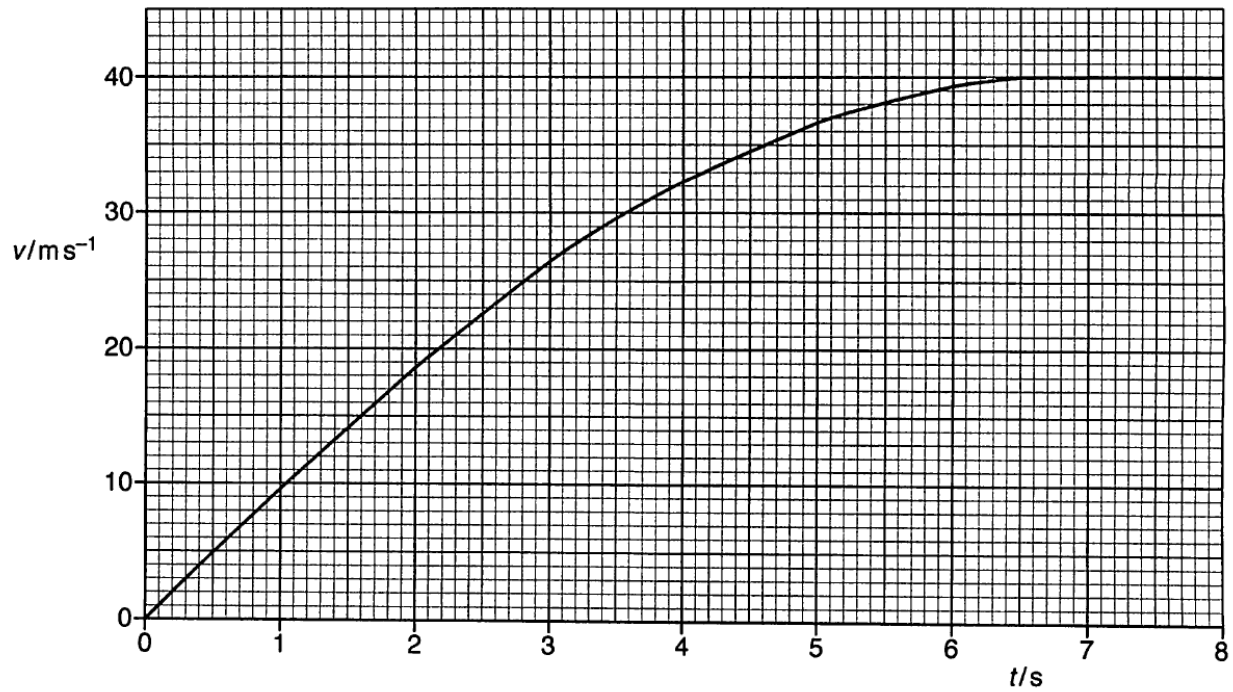
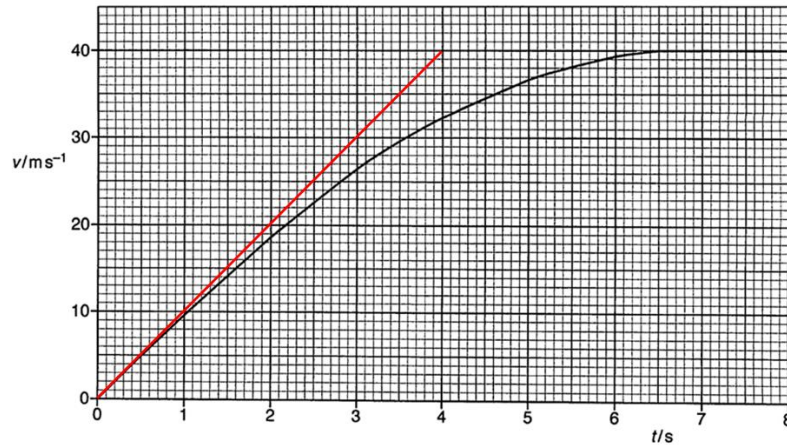


Fig 1.1

- (a) (i) Use Fig 1.1. to determine the acceleration of the ball at time $t = 0$. Show your construction on Fig 1.1.

acceleration = m s^{-2} [3]

Solution:



acceleration
 = gradient of the $v - t$ graph at $t = 0$
 $= \frac{40 - 0}{4 - 0}$
 $= 10 \text{ m s}^{-2}$

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1
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(ii) By reference to your answer in (a)(i), suggest the difference, if any, between your answer and the acceleration of free fall.

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 [2]

Solution:

The value obtained in (a)(i) is close to the value of the acceleration of the free fall.

At time $t = 0$, the velocity of the ball is (close to) 0, so no (or very little) drag force acting on it. The net force = weight, acceleration = g

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(b) Calculate the maximum resistive force acting on the ball.

force = N [1]

Solution:

Maximum resistive force acts when the ball is at terminal velocity ($a = 0$)

At this point,

$$F_{\text{net}} = mg - F_{\text{resistive}} = 0$$

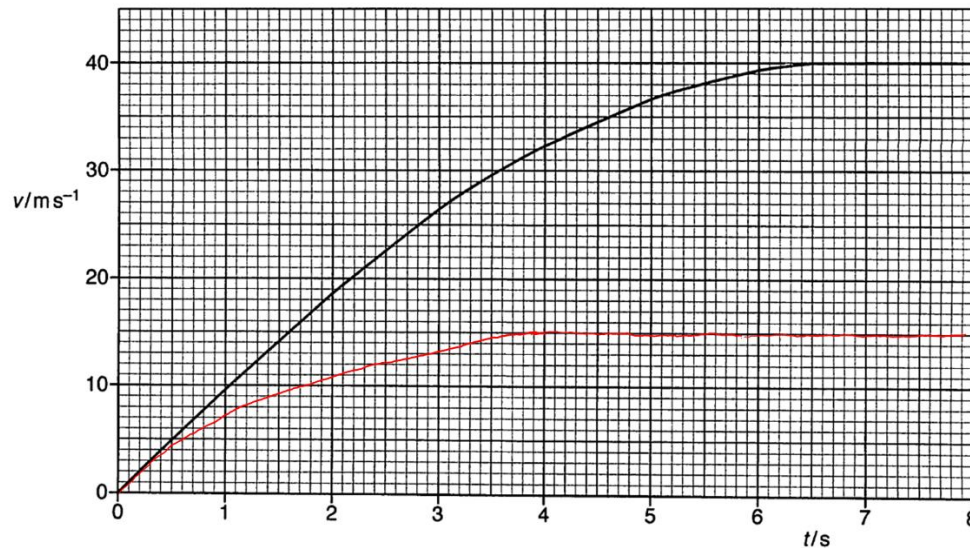
$$F_{\text{resistive}} = mg = 0.01 \times 9.81 = 0.0981 \text{ N}$$

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- (c) On Fig 1.1, draw another curve to show the variation with time t of the speed of the ball v if the ball was dropped in a more viscous medium. [2]

[Total: 8]

Solution:



1 mark – rate of increase of v is lower than the original graph.

Note: $mg - kv = ma$, k is larger and hence acc is smaller

1 mark – object reaches terminal velocity at a lower velocity and at an earlier time.

Note: When $mg = kv$, terminal vel is reached or when $a = 0$, Hence acc = 0 is reached earlier when k is larger.